

AI-BASED SMART ATTENDANCE SYSTEM USING FACE RECOGNITION IN CLASSROOM

A PROJECT REPORT

Submitted by

AARTHI N (810022104079)

BHAVATHARINI S (810022104090)

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TIRUCHIRAPPALLI – 620 024

ANNA UNIVERSITY : CHENNAI 600 025

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ANNA UNIVERSITY : CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report “ **AI-BASED SMART ATTENDANCE USING FACE RECOGNITION IN CLASSROOM**”, is the Bonafide work of **AARTHI N (810022104079) and BHAVATHARINI S (810022104090)** who carried out the project under my supervision.

SIGNATURE

Dr. G. ANNAPOORANI

HEAD OF THE DEPARTMENT

Assistant Professor(Sl.Grade),
Computer Science & Engineering,
BIT Campus, Anna University
Tiruchirappalli – 620 024.

SIGNATURE

Dr. V. M. PRIYADHARSHINI

SUPERVISOR

Assistant Professor(Sl.Grade),
Computer Science & Engineering
BIT Campus, Anna University
Tiruchirappalli – 620 024.

Certified that **AARTHI N and BHAVATHARINI S** were examined in a Viva –
Voce examination held on _____

Internal Examiner

External Examiner

DECLARATION

We hereby declare that the work entitled “**AI-BASED SMART ATTENDANCE USING FACE RECOGNITION IN CLASSROOM** ” is submitted in partial fulfillment of the requirement for the award of the degree in B.E, in University College of engineering, BIT Campus, Anna University, Tiruchirappalli. It is the record of my own work carried out during the academic year 2025 – 2026 under the supervision and guidance of **DR.V.M.PRIYADHARSHINI**, Assistant Professor(Sl.Grade), Department of Computer Science and Engineering, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli. The extent and source of information are derived from the existing literature and have been indicated through the dissertation at the appropriate places. The matter embodied in this work is original and has not been submitted for the award of any other degree, either in this or any other university.

AARTHI N (810022104079)

BHAVATHARINI S (810022104090)

I Certify that the declaration made above by the candidate is true.

Signature of the Guide

DR.V.M.PRIYADHARSHINI,

Assistant Professor (Sl.Grade),

Department of CSE,

University College of Engineering,

BIT Campus, Anna University,

Tiruchirappalli – 620024 .

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ABSTRACT

The increasing demand for automation in educational institutions and organizations has led to the development of intelligent systems that improve efficiency and accuracy in routine tasks. Attendance management is one such essential activity that is traditionally performed using manual or semi-automated methods, which are often time-consuming, error-prone, and vulnerable to proxy attendance. To overcome these limitations, this project proposes an AI-Based Smart Attendance System using Face Recognition. The system utilizes advanced computer vision and deep learning techniques to automatically detect and recognize individuals in real time. Face detection is performed using RetinaFace, a robust deep learning-based model capable of accurately identifying faces under varying conditions such as illumination, pose, and occlusion. For face recognition, the system employs InsightFace with ArcFace, which generates highly discriminative facial embeddings and enables precise identity verification. The system captures live video input through a webcam, processes frames to detect faces, and extracts facial features to generate embeddings. These embeddings are compared with stored data using similarity measures such as cosine similarity and Euclidean distance to identify individuals. Once recognized, attendance is automatically recorded and stored in a database with time and session details. To improve reliability, the system validates attendance across multiple checkpoints during a session, ensuring that presence is consistent rather than momentary. This reduces the chances of false attendance marking and enhances system accuracy. The implementation is carried out using Python, along with libraries such as OpenCV, NumPy, and deep learning frameworks. The proposed system demonstrates high accuracy efficient real-time performance, and reduced manual workload. It provides a secure, contactless, and scalable solution for attendance management. This system can be effectively deployed in educational institutions, workplaces, and smart environments, contributing to digital transformation and intelligent automation.

சுருக்கம்

கல்வி மற்றும் நிறுவனங்களில் தானியக்கத்தின் தேவைகள் அதிகரித்து வருவதால், வருகைப் பதிவேடு மேலாண்மை போன்ற செயல்களை துல்லியமாகவும் விரைவாகவும் செய்ய நுண்ணறிவு அமைப்புகள் பயன்படுத்தப்படுகின்றன. இதற்காக, முகஅடையாளத்தை பயன்படுத்தும் செயற்கை நுண்ணறிவு அடிப்படையிலான ஸ்மார்ட் வருகைப் பதிவு அமைப்பு முன்மொழியப்படுகிறது.

இந்த அமைப்பு RetinaFace மூலம் முகத்தை கண்டறிந்து, InsightFace (ArcFace) மூலம் அடையாளம் காண்கிறது. வெப்கேம் மூலம் பெறப்படும் வீடியோவில் இருந்து முக அம்சங்கள் எடுக்கப்பட்டு, சேமிக்கப்பட்ட தரவுகளுடன் ஒப்பிடப்படுகிறது. சரியான அடையாளம் கண்டறியப்பட்டவுடன், வருகை தானாகவே பதிவு செய்யப்படுகிறது.

அமைப்பின் துல்லியத்தை உயர்த்த, பல கட்டங்களில் வருகை சரிபார்க்கப்படுகிறது. Python, OpenCV, NumPy போன்றவற்றை பயன்படுத்தி உருவாக்கப்பட்ட இந்த அமைப்பு 90% க்கும் மேற்பட்ட துல்லியத்துடன் செயல்படுகிறது. இது பாதுகாப்பான மற்றும் செயல்திறன் மிக்க தீர்வாக கல்வி மற்றும் அலுவலக சூழல்களில் பயன்படுத்தப்படலாம்.

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LIST OF ABBREVIATIONS & ACRONYMS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
CV	Computer Vision
DL	Deep Learning
FR	Face Recognition
DB	Database
GUI	Graphical User Interface
API	Application Programming Interface
CSV	Comma Separated Values
FPS	Frames Per Second
LBPH	Local Binary Pattern Histogram
UI	User Interface
ReLU	Rectified Linear Unit
FD	Face Detection
COS	Cosine Similarity
CPU	Central Processing Unit
RAM	Random Access Memory
OS	Operating System
IDE	Integrated Development Environment

CHAPTER 1

INTRODUCTION

In recent years, the rapid advancement of Artificial Intelligence (AI) and Computer Vision has significantly influenced the development of intelligent automation systems. One such critical application is the automation of attendance management systems in educational institutions and organizations. Traditional attendance systems, which rely on manual entry or physical interaction, are increasingly becoming inefficient, error-prone, and vulnerable to manipulation.

The proposed project, “AI-Based Smart Attendance System Using Face Recognition in classroom”, aims to address these challenges by leveraging deep learning-based facial recognition techniques to automate and secure the attendance process. Face recognition is a biometric technology that identifies or verifies individuals based on their facial features, which are unique and difficult to replicate.

Unlike traditional biometric systems such as fingerprint or RFID-based attendance, face recognition systems operate in a non-contact and unobtrusive manner, making them highly suitable for modern environments. The integration of real-time image processing, deep neural networks, and intelligent decision-making algorithms enables the system to accurately detect and recognize multiple individuals simultaneously.

This project utilizes advanced techniques such as: RetinaFace-based face detection, InsightFace (ArcFace)-based feature extraction and recognition, Cosine similarity for identity verification, Real-time video stream processing.

RetinaFace is a state-of-the-art deep learning-based face detector that provides highly accurate face localization along with facial landmark detection, making it robust under varying lighting conditions and pose. InsightFace, powered by the ArcFace loss function, generates highly discriminative facial embeddings that improve recognition accuracy by enforcing angular separation between different

identities.

These technologies collectively contribute to a robust, scalable, and highly accurate attendance system, suitable for real-world deployment.

1.1 Background

The academic environment demands timely and trustworthy attendance collection. Faculty members often begin a class by taking attendance through a physical register, by calling out student names, or by asking students to sign a sheet. Although these practices are simple to initiate, they become inefficient when the class strength is high. A classroom of sixty students may require several minutes for accurate roll call, particularly when the class includes late arrivals or name ambiguities.

1.1.1 Context of the Problem

Attendance management is an essential administrative activity in educational institutions and organizations. Traditionally, this process has been carried out manually or through semi-automated systems, which often lack efficiency and reliability. As the number of students increases, managing attendance becomes more complex, time-consuming, and prone to inaccuracies.

Manual attendance methods are particularly inefficient in large classrooms, where significant time is spent recording attendance. Moreover, these systems are highly vulnerable to manipulation, such as proxy attendance, where one individual marks attendance on behalf of another. This reduces the credibility and accuracy of attendance records.

With the growing demand for automation and digital transformation, there is a need for intelligent systems that can handle attendance management efficiently while ensuring accuracy, security, and scalability.

1.2 Problem Statement

Attendance management remains a critical yet challenging task in educational institutions and organizations. Traditional and semi-automated attendance systems are unable to meet the growing demand for accuracy, efficiency, and security in

large-scale environments.

Despite the availability of various technologies, there is still a lack of a unified system that can perform real-time, automated, and reliable attendance marking without human intervention. Existing methods often fail to ensure authenticity and are vulnerable to manipulation, thereby affecting the integrity of attendance records.

The need of the hour is to develop an intelligent system that can accurately identify individuals in real time while eliminating manual effort and preventing fraudulent practices.

1.3 Objectives

The objective of this project is to design and develop an AI-based smart attendance system that automatically detects and recognizes individuals using facial features and records their attendance in real time.

The system aims to utilize advanced deep learning models, specifically RetinaFace for accurate face detection and InsightFace (ArcFace) for robust face recognition, to ensure high accuracy, reliability, and security. The objective is to eliminate manual attendance processes, prevent proxy attendance, and enable efficient real-time monitoring in educational and organizational environments.

The specific objectives of the proposed system are defined in a sequential and structured manner to achieve the overall goal effectively.

The system is designed to operate in real time with minimal latency while maintaining scalability, allowing it to be deployed effectively in real-world environments such as educational institutions and offices.

1.4 Scope of the Project

The scope of this project involves the development of an intelligent and automated attendance management system using advanced face recognition techniques. The system is designed to operate in real-time environments, where live video input is captured through a webcam and processed continuously to detect and recognize individuals. By integrating RetinaFace for face detection and

InsightFace (ArcFace) for face recognition, the system ensures high accuracy and robustness under various real-world conditions.

The project primarily focuses on enabling automatic attendance marking without manual intervention. Once a face is detected and successfully recognized, the system records the attendance along with relevant details such as identity and timestamp. The system is also capable of handling multiple individuals simultaneously, making it suitable for classroom and group scenarios.

In addition, the project includes the implementation of an efficient data storage mechanism using lightweight databases such as JSON or SQLite, ensuring effective management of user data and attendance records. The system is designed for practical deployment in environments such as educational institutions, offices, and laboratories, where automated attendance tracking is essential.

However, the scope of the project is subject to certain limitations. The performance of the system may be affected by environmental factors such as lighting conditions, camera quality, and background variations.

Chapter 2

Literature Review

2.1 Introduction

The literature review plays a vital role in understanding the current state of research and technological advancements related to attendance management systems and face recognition. It involves a systematic study of previously developed systems, methodologies, and approaches that have been proposed to improve the efficiency, accuracy, and security of attendance tracking processes.

Attendance management has evolved significantly over the years, beginning with manual methods and gradually transitioning to semi-automated and fully automated systems. Traditional approaches such as manual attendance registers are simple but highly inefficient and prone to human error. To overcome these issues, technologies such as RFID and biometric systems were introduced. While these systems improved speed and reliability to some extent, they still suffer from limitations such as dependency on physical devices, susceptibility to proxy attendance, and lack of scalability in large environments.

With the rapid growth of Artificial Intelligence and Computer Vision, more advanced solutions have emerged in the form of face recognition-based systems. These systems utilize deep learning algorithms to automatically detect and identify individuals based on their facial features. The use of Convolutional Neural Networks (CNNs) has significantly enhanced the ability of these systems to perform under complex real-world conditions, including variations in lighting, pose, and background.

Recent developments in this domain have introduced highly efficient models such as RetinaFace for face detection, which provides accurate localization of faces along with facial landmarks, and InsightFace (ArcFace) for face recognition, which generates highly discriminative embeddings for precise identification. These advancements have enabled the development of robust, real-time attendance systems capable of handling multiple individuals simultaneously.

In this chapter, a detailed analysis of five existing attendance systems is presented based on the provided research papers. Each system is examined in terms of its methodology, technologies used, performance, advantages, and limitations. This comparative study helps in identifying the gaps in existing solutions and highlights the need for a more efficient and intelligent system.

Furthermore, the insights gained from the literature review form the foundation for the proposed system. By addressing the shortcomings of existing approaches, the proposed solution aims to develop a real-time, accurate, and secure attendance system using advanced deep learning techniques.

2.2 Review of Existing Systems and Papers

TITLE: SMART ATTENDANCE AND ATTENTION MONITORING SYSTEM (2023)

AUTHOR: SHILPA MANGESH PANDE, SUBHASSINI SRIDHARAN, SWASTIK RAJ SINGH.

This paper proposes a smart attendance system that integrates face recognition with student attention monitoring. The system uses a camera to capture classroom images and detect student faces using the YOLO (You Only Look Once) algorithm. After detection, the system compares the faces with a trained dataset to identify students and mark attendance automatically.

In addition to attendance marking, the system also analyzes student attentiveness using parameters such as Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and gaze angle. This allows the system to monitor whether students are actively engaged during lectures, thereby enhancing the overall learning process.

Limitations:

Despite its advanced features, the system requires high computational resources due to real-time video processing and attention analysis. The accuracy of the system may be affected by variations in lighting conditions and facial orientations. Additionally, continuous monitoring of attention may raise privacy

concerns, and the system may not perform efficiently in highly crowded classrooms.

TITLE: SMART ATTENDANCE REGISTRATION AND MANAGEMENT SYSTEM THROUGH AI (2023)

AUTHOR(S): G. RAJESH, N. SAIRAM, M. PRAVEEN

This paper presents an AI-based attendance system that utilizes Automatic Facial Recognition (AFR) to track student attendance in real time. The system captures facial images from video input and compares them with stored database images to identify students. It uses deep learning techniques and a modified Convolutional Neural Network (CNN) architecture, specifically incorporating a Split Invariant Artificial Neural Network (SIANN) model to improve recognition accuracy and robustness against variations in facial expressions and environmental conditions.

The system also includes features such as attendance logging, report generation, and real-time monitoring. Once a match is found, attendance is recorded automatically along with timestamps, and detailed reports can be generated for administrative use. This enhances transparency and reduces manual workload in attendance management systems.

Limitations:

The system requires large datasets for training to achieve high accuracy, which increases preprocessing time and storage requirements. It is computationally intensive and may not perform efficiently on low-end hardware systems, making real-time deployment challenging in resource-constrained environments. Additionally, performance may degrade under challenging conditions such as occlusions, poor lighting, facial blur, or extreme facial variations.

TITLE: AUTOMATED ATTENDANCE SYSTEM USING FACE RECOGNITION(2023) (OPENCV & DLIB BASED)

AUTHOR(S): ASHESH KUMAR, SUMITRA SAMAL, MANPREET SINGH SALUJA, AARYAN TIWARI

This paper proposes an automated attendance system using face recognition techniques implemented with OpenCV and Dlib libraries. The system captures images through a camera and detects faces in real time using classical computer vision techniques. The detected faces are then compared with stored images in the database, and attendance is marked automatically for recognized individuals.

The system supports multi-face detection and provides quick response time, making it suitable for classroom environments. It also includes database integration for storing attendance records and generating reports, which improves data management and accessibility. The approach is relatively simple and does not require heavy computational infrastructure, making it suitable for basic implementations.

Limitations:

The system is limited in handling large variations in facial angles and typically recognizes faces within a restricted angle range (around 30 degrees). It is also highly sensitive to lighting conditions, shadows, and background noise, which can significantly affect recognition accuracy. Furthermore, the system lacks robustness in real-time high-density environments and requires improvements in recognition speed, scalability, and adaptability for real-world deployment.

TITLE: WI-FI BASED SMART ATTENDANCE MONITORING SYSTEM (2023)

AUTHOR(S): NITHIN RAMAKRISHNAN, AJIL A, ANOOJA ALI, MEENAKSHI SUNDARAM, SANJU V.

This system combines Wi-Fi-based attendance tracking with face recognition techniques to provide a mobile-enabled smart attendance solution. It allows users

to mark attendance through a mobile application while the system performs identity verification using facial recognition. The system includes features such as login authentication, subject management, attendance tracking, report generation, and communication modules between students and staff.

Face recognition is used as an additional verification layer, where the system captures images using a mobile camera and compares them with stored data to validate identity before marking attendance. It also provides notifications and reminders for attendance marking and enables administrators to remotely monitor attendance records, improving accessibility and management efficiency.

Limitations:

The system heavily depends on stable network connectivity, and performance may degrade in low or unstable Wi-Fi conditions. The reliance on mobile devices introduces variability in camera quality, processing speed, and user experience. Additionally, security concerns may arise due to login-based access mechanisms, and the system may not be fully reliable in large-scale or real-time classroom environments where simultaneous processing is required.

TITLE: CLOUD BASED USER INTERFACE DESIGN FOR SMART STUDENT ATTENDANCE SYSTEM

AUTHOR(S): TRIO ADIONO, DIDI SETIAWAN, MAURIZFA, JASON WILLIAM AND NANA SUTISNA

This paper focuses on an automated attendance system using traditional image processing techniques combined with basic face recognition algorithms. The system uses Haar Cascade classifiers for face detection and applies feature extraction techniques to identify individuals. The recognition process involves converting images into grayscale, extracting facial features, and comparing them with stored templates in the database.

The system automates attendance marking and stores records in a centralized database, which can later be used for reporting and performance analysis. It is relatively simple to implement and requires fewer computational resources

compared to deep learning-based models, making it suitable for small-scale applications.

Limitations:

The use of traditional image processing techniques results in lower accuracy compared to modern deep learning-based systems. The system is highly sensitive to environmental conditions such as lighting variations, shadows, and background complexity. It also struggles with real-time processing, multi-face detection, and large-scale deployments, limiting its effectiveness in practical classroom scenarios.

2.3 Comparative Analysis

Table 1. Comparative Analysis

Paper No.	System	Methods	Accuracy	Tools
1.	Smart Attendance and Attention Monitoring System	Face detection with attention monitoring system using object detection	90% – 94%	YOLO, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Computer Vision
2.	Smart Attendance Registration and Management System through AI	Deep learning-based facial recognition system	95% – 98%	CNN, SIANN Model, Automatic Face Recognition (AFR)

3.	Automated Attendance System Based on Face Recognition Using Opencv	Face detection and recognition using image processing techniques	80% – 88%	OpenCV, Dlib, Image Processing Algorithms
4.	Wi-Fi Based Smart Attendance Monitoring System	Mobile and Wi-Fi based attendance system with face verification	85% – 92%	Mobile Application, Wi-Fi Network, Face Recognition Module
5.	Cloud Based User Interface Design for Smart Student Attendance System	Traditional image processing-based face recognition system	75% – 85%	Haar Cascade Classifier, Feature Extraction, OpenCV

Chapter 3

Proposed System

3.1 Existing System

The existing attendance systems used in educational institutions include manual attendance registers, token-based systems (RFID, ID cards, barcodes), contact biometric systems, and basic software-based attendance tools. These methods are widely used due to their simplicity and availability.

Manual systems rely on paper records, while token-based systems use physical cards for identification. Biometric systems use fingerprint scanning for identity verification, and software-based systems allow attendance marking through digital interfaces. Although these systems provide basic functionality, they have limitations in efficiency, accuracy, and security.

3.1.1 Working of Existing Systems

Existing attendance management systems operate based on predefined identification and verification mechanisms, which typically follow a sequential workflow. In traditional manual systems, attendance is recorded by calling out names or marking entries in a register, where the process is entirely dependent on human involvement. This approach is simple but lacks efficiency and accuracy, especially in large-scale environments.

In more advanced systems, technologies such as RFID and biometric authentication are used to automate the identification process. In RFID-based systems, each user is assigned a unique identification card embedded with a radio frequency tag. The attendance is recorded when the user scans the card using a reader device, which retrieves the stored information and updates the attendance database.

Biometric systems, such as fingerprint or iris recognition, further enhance the identification process by using unique physiological characteristics of individuals. In these systems, the user places their finger or eye on a scanning

device, and the captured data is compared with stored templates to verify identity before marking attendance.

Despite differences in implementation, most existing systems follow a similar workflow consisting of three main stages: user identification, verification using stored data, and attendance recording. These systems are designed to reduce manual effort; however, they still rely on user interaction and external devices for operation.

3.1.2 Limitations of Existing Systems

Although existing attendance systems provide partial automation, they suffer from several critical limitations that restrict their effectiveness in real-world scenarios.

One of the major drawbacks is the dependency on manual intervention or user participation. Even in semi-automated systems such as RFID or biometric-based methods, users are required to actively scan cards or provide biometric input, which slows down the process and reduces efficiency in large groups.

Another significant issue is the vulnerability to proxy attendance. In systems such as RFID or manual registers, individuals can easily manipulate the process by marking attendance on behalf of others. This compromises the authenticity and reliability of attendance records.

Hardware dependency is also a major limitation. RFID readers, biometric scanners, and other specialized devices increase the overall cost of the system and require regular maintenance. Any hardware failure can disrupt the entire attendance process.

Time inefficiency is another concern, particularly in environments with a large number of users. Since many systems process individuals sequentially, the overall time required to record attendance increases significantly.

Additionally, contact-based biometric systems raise hygiene concerns, as multiple users are required to interact with the same device. This makes such

systems less suitable in modern environments where contactless solutions are preferred.

Finally, existing systems lack scalability and adaptability. They are often difficult to extend or modify for larger datasets or different environments, limiting their practical usability.

3.2 Proposed System

The proposed system is an AI-based smart attendance system that uses face recognition technology to automate the attendance process in classrooms and organizations. It is designed to improve accuracy, reduce manual effort, and eliminate issues such as proxy attendance. The system captures live video input and identifies individuals automatically without requiring any physical interaction.

Unlike traditional methods, this system relies on advanced deep learning techniques to ensure reliable performance even under different environmental conditions. It provides a contactless, efficient, and secure solution for attendance management, making it suitable for modern smart environments.

3.2.1 Overview of the Proposed System

The proposed system is an AI-based smart attendance system using Face Recognition in Classroom designed to automate the process of attendance marking using advanced face recognition techniques. The system operates in real time by capturing live video input through a camera, detecting human faces, extracting unique facial features, and identifying individuals by comparing them with a pre-registered database.

Unlike traditional attendance systems, the proposed solution eliminates the need for manual intervention or physical interaction. It leverages deep learning models to ensure accurate and reliable identification of individuals even under varying environmental conditions such as changes in lighting, pose variations, and background complexity.

The system utilizes RetinaFace for precise face detection, which is capable of identifying faces along with facial landmarks, and InsightFace (ArcFace) for feature extraction and recognition, which generates highly discriminative embedding vectors for each individual. These embeddings are compared using cosine similarity to determine identity.

Once a match is found, the system automatically records attendance along with relevant details such as user identity, date, and timestamp. The entire process is performed in real time, making the system efficient, scalable, and suitable for practical deployment in classrooms and organizational environments.

3.2.2 System Concept

The core concept of the proposed system is to create a fully automated, contactless attendance mechanism using face recognition technology. The system continuously monitors a video stream, processes each frame, and performs the following operations: Captures real-time video input from a webcam, Detects faces using RetinaFace, Aligns and preprocesses detected faces, Extracts facial features using InsightFace (ArcFace), Compares extracted features with stored embeddings, Identifies individuals based on similarity score, Automatically marks attendance

This pipeline ensures that the attendance process is accurate, efficient, and free from manual errors or fraudulent practices such as proxy attendance.

3.2.3 Key Features

The system supports real-time face detection and recognition, allowing it to process live video streams efficiently without noticeable delay. This ensures immediate attendance marking and continuous monitoring throughout the session. The real-time capability enhances system responsiveness and provides a seamless user experience without interruptions.

It is capable of multi-face processing, meaning it can detect and recognize multiple individuals simultaneously within a single frame. This makes it suitable for classroom environments where many students are present at the same time.

The system efficiently handles group scenarios and maintains performance even when the number of users increases.

The system provides automatic attendance logging, eliminating the need for manual entry and reducing the chances of human error. Attendance is recorded instantly once an individual is recognized, along with essential details such as date, time, and session information. This improves accuracy and ensures reliable record maintenance.

A secure face-based registration mechanism is implemented to store user data in the form of embeddings, ensuring reliable and consistent identification during future sessions. This approach enhances security by preventing duplication and unauthorized access, as each face is uniquely represented.

Additionally, the system includes an optional gender classification feature, which demonstrates extensibility and the ability to integrate additional AI-based functionalities. This shows that the system can be further enhanced with more intelligent features in the future without major changes to the core design.

The use of a lightweight storage mechanism ensures efficient data handling with minimal computational overhead, making the system suitable for deployment on standard hardware. It reduces the need for complex infrastructure while still maintaining performance and reliability.

It provides a fully automated and contactless solution, eliminating manual effort and improving user convenience. This also promotes better hygiene and safety by avoiding physical contact with devices. The use of advanced deep learning models ensures high accuracy and reliability, even in challenging real-world conditions such as varying lighting, facial expressions, and orientations.

The system effectively prevents proxy attendance, as identification is based on unique facial features that cannot be easily replicated. It also supports real-time processing, enabling quick and efficient attendance marking without delays, even during peak usage.

Furthermore, the system is scalable and adaptable, allowing it to be extended for use in larger environments or integrated with other systems such as cloud platforms and mobile applications. This flexibility ensures long-term usability and makes the system suitable for future technological advancements.

3.3 System Architecture

3.3.1 Block Diagram



Figure 1. Block Diagram

3.3.2 Explanation of Modules

Image Acquisition Module

The image acquisition module is responsible for capturing real-time video input through a webcam. It continuously reads frames from the video stream and converts them into a format suitable for further processing.

This module ensures a continuous flow of data, enabling the system to operate in real time without interruptions. Each captured frame is passed to the face detection module for further analysis.

Face Detection Module (RetinaFace)

The face detection module uses the RetinaFace model, which is a deep learning-based face detector known for its high accuracy and robustness.

RetinaFace not only detects the presence of faces but also identifies facial landmarks such as eyes, nose, and mouth. This helps in improving alignment and recognition accuracy.

Key capabilities: Detects multiple faces simultaneously ,Works under varying lighting conditions, Handles pose variations and occlusions

Face Preprocessing and Alignment Module

Once faces are detected, they are passed to the preprocessing module. In this stage: The face region is extracted (cropped), The image is resized to a standard dimension (e.g., 112×112 or 160×160), Pixel values are normalized, Facial alignment is performed using detected landmarks

This step ensures consistency in input data, which improves the performance of the recognition model.

Feature Extraction Module (InsightFace – ArcFace)

This is the most critical module of the system. It uses the InsightFace framework, which is based on the ArcFace loss function, to extract facial features.

The model processes the input face and generates a 512-dimensional embedding vector, which uniquely represents the individual.

Key characteristics: Deep CNN-based architecture (ResNet backbone), High discriminative power, Robust against variations in lighting and pose

These embeddings serve as the basis for identifying individuals.

Recognition Module (Similarity Computation)

In this module, the facial embedding generated by the InsightFace (ArcFace) model is compared with the stored embeddings in the database to identify the individual. The comparison is performed using cosine similarity, which measures how similar two vectors are.

Formula: $\cos(\theta) = \frac{A \cdot B}{||A|| ||B||}$

Where:

A = embedding vector of the input face

B = embedding vector stored in the database

A·B= dot product of the vectors

$\|A\|, \|B\|$ = magnitude (length) of the vectors

Explanation

Cosine similarity calculates the angle between two embedding vectors. These vectors represent facial features extracted by the InsightFace model.

If the angle between the vectors is small, the similarity value will be close to 1, indicating that both faces belong to the same person.

If the angle is large, the similarity value will be closer to 0, indicating that the faces are different.

Interpretation

Value close to 1 → High similarity (same person) & Value close to 0 → Low similarity (different person)

Decision-Making Module

Based on the similarity score obtained from the recognition module, the system makes a decision:

If similarity $\geq$ predefined threshold → Face is recognized

If similarity $<$ threshold → Face is classified as unknown

This threshold-based decision ensures reliable identification while minimizing false positives.

Attendance Management Module

Once a face is recognized, this module automatically records attendance. It logs: User identity, Date, Time

The system ensures that attendance is recorded only once per session to avoid duplication.

Database Module

The database module stores: User details, Facial embeddings, Attendance records.

A lightweight database such as SQLite or JSON is used to ensure efficient storage and fast retrieval.

3.4 Methodology

The proposed system follows a deep learning-based pipeline for automated attendance marking. The methodology involves capturing real-time video input, detecting faces, extracting unique facial features, recognizing individuals, and recording attendance.

3.4.1 Step-by-Step Working of the System

Step 1: System Initialization

The system is initialized by loading the trained models (RetinaFace and InsightFace) and connecting to the database containing stored facial embeddings and user information.

Step 2: Image Acquisition

A webcam is activated to capture real-time video input. The video stream is divided into frames, which are processed continuously.

Step 3: Face Detection (RetinaFace Algorithm)

Each frame is passed to the RetinaFace model, which detects faces and identifies facial landmarks such as eyes, nose, and mouth. Detects multiple faces in a single frame, Provides bounding box coordinates, Ensures high detection accuracy.

Step 4: Face Preprocessing and Alignment

The detected face regions are extracted and preprocessed: Cropped from the original frame, Resized to standard dimensions, Normalized pixel values, Aligned using facial landmarks

Step 5: Feature Extraction (InsightFace – ArcFace Algorithm)

The preprocessed face is passed to the InsightFace model, which uses the ArcFace algorithm to generate a 512-dimensional embedding vector.

This embedding uniquely represents the individual's facial features.

Step 6: Face Recognition (Cosine Similarity Algorithm)

The generated embedding is compared with stored embeddings in the database using cosine similarity. Similarity score is calculated, Higher score indicates closer match

Step 7: Decision Making

A threshold value is used to determine identity: If similarity $\geq$ threshold $\rightarrow$ Face is recognized, If similarity $<$ threshold $\rightarrow$ Face is marked as unknown

Step 8: Attendance Marking

If the face is recognized: Attendance is marked automatically, Name, date, and time are recorded, Duplicate entries are prevented

Step 9: Output Display

The system displays the result on the screen: Recognized person's name, Unknown label if not matched, Bounding box around detected faces

Registration Algorithm

Before recognition, users must be registered in the system.

Steps:

1. Capture multiple face images using webcam
2. Detect face using RetinaFace
3. Preprocess and align the face
4. Generate embedding using InsightFace
5. Store embedding in database
6. Associate embedding with user identity

Attendance Marking Algorithm

Steps:

1. Receive recognition result
2. Check if user is already marked present
3. If not marked $\rightarrow$ record attendance
4. Store timestamp and user details
5. Prevent duplicate entries

3.4.2 Flowchart of the Proposed System

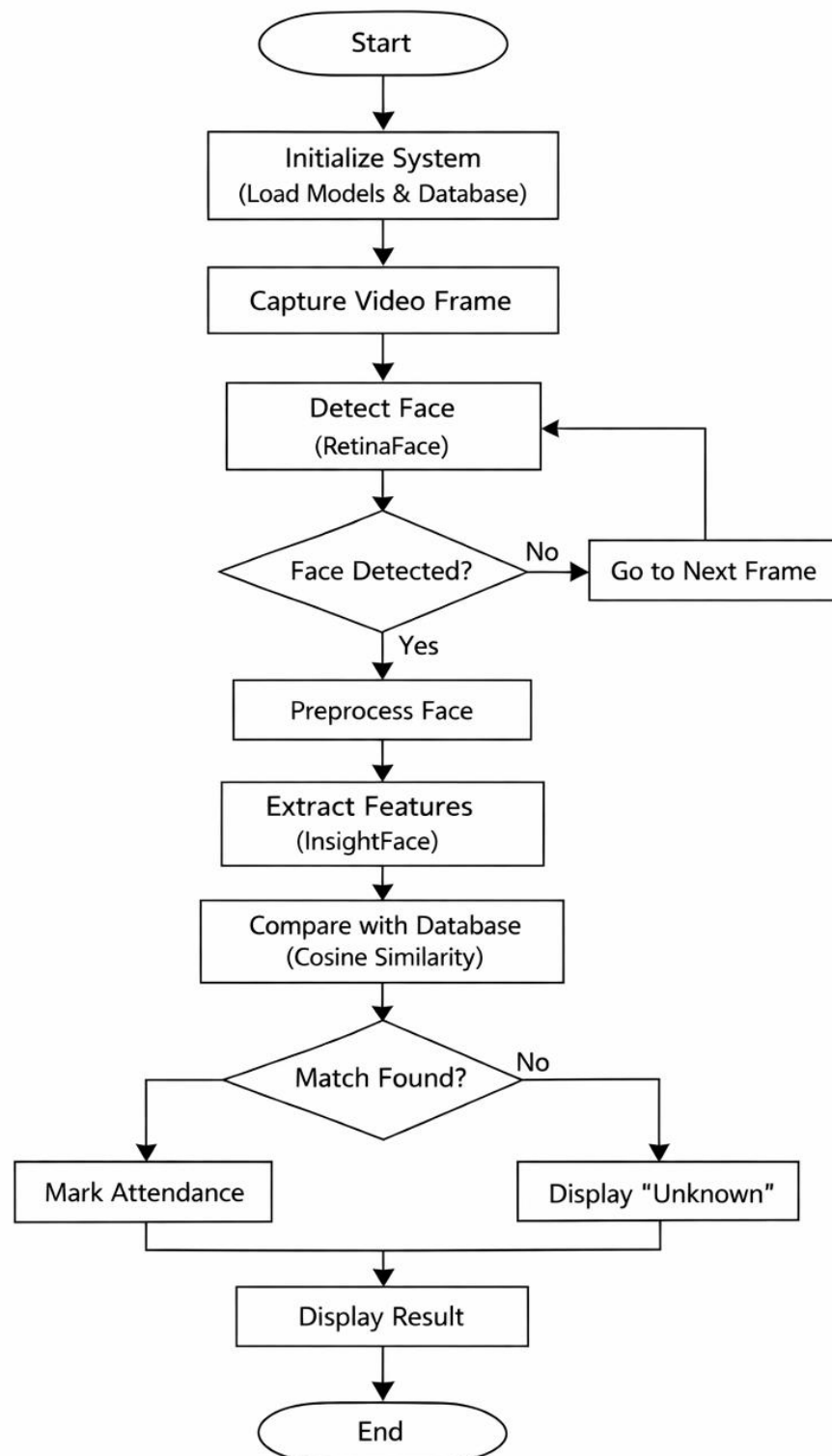


Figure 2. Flowchart

3.5 Module Description

The proposed system is designed using a modular architecture, where each module performs a specific task in the overall attendance process. This modular design improves system clarity, maintainability, and scalability.

Module 1: Image Acquisition Module

The Image Acquisition Module is responsible for capturing real-time video input using a webcam. It continuously reads frames from the video stream and converts them into a format suitable for further processing.

This module ensures a steady and uninterrupted flow of image data to the system. Each captured frame is forwarded to the face detection module for analysis. The performance of the entire system depends on the quality and consistency of input provided by this module.

Module 2: Face Detection Module (RetinaFace)

The Face Detection Module uses the RetinaFace algorithm, which is a deep learning-based model designed for accurate face detection.

This module identifies the presence of faces in each frame and provides bounding box coordinates along with facial landmarks such as eyes, nose, and mouth. These landmarks are essential for proper alignment and preprocessing.

The module is capable of detecting multiple faces simultaneously and performs well under different lighting conditions and orientations, making it suitable for real-time applications.

Module 3: Face Preprocessing and Alignment Module

After detection, the face regions are extracted and passed to the preprocessing module. This module prepares the detected faces for feature extraction.

The preprocessing steps include cropping the face region, resizing it to a standard dimension (such as 112×112), normalizing pixel values, and aligning the face using detected landmarks.

This process ensures that all input images are consistent, which improves the

accuracy and performance of the recognition model.

Module 4: Feature Extraction Module (InsightFace – ArcFace)

This module uses the InsightFace framework, based on the ArcFace algorithm, to extract unique facial features.

The processed face image is converted into a 512-dimensional embedding vector, which acts as a numerical representation of the individual's facial characteristics. These embeddings are highly discriminative and help distinguish between different individuals accurately.

This module plays a crucial role in ensuring high recognition accuracy.

Module 5: Face Recognition Module

The Face Recognition Module compares the generated embedding with stored embeddings in the database using cosine similarity.

It calculates a similarity score to determine whether the detected face matches a registered user. Based on the similarity value, the system identifies the individual or classifies the face as unknown.

This module ensures reliable identity verification and minimizes false matches.

Module 6: Attendance Management Module

The Attendance Management Module is responsible for recording attendance automatically after successful recognition.

It logs important details such as the user's name, date, and time of attendance. The module also ensures that duplicate entries are avoided by checking whether attendance has already been marked for a particular session.

This module eliminates manual effort and ensures accurate record-keeping.

Module 7: Database Module

The Database Module stores all system data, including user details, facial embeddings, and attendance records.

A lightweight database system such as SQLite or JSON is used to ensure efficient storage and quick data retrieval. The module supports both read and write operations and plays a key role in maintaining system data integrity.

Module 8: User Interface Module

The User Interface Module provides interaction between the user and the system. It displays the live video feed along with detection results such as bounding boxes and recognized names.

This module also allows users to perform operations such as registration and viewing attendance records. It ensures that the system is user-friendly and easy to operate.

Chapter 4

Implementation & Results

4.1 Tools & Technologies

The system is developed using Python, which provides strong support for artificial intelligence and computer vision. OpenCV is used for real-time image processing and face detection, while NumPy handles numerical operations efficiently.

Face recognition is implemented using advanced models like InsightFace and RetinaFace for accurate detection and feature extraction. SQLite is used as a lightweight database to store student data and attendance records.

Overall, the system uses efficient tools and technologies that ensure real-time performance and easy deployment on standard hardware.

4.1.1 Software Requirements

Table 2. Software Requirements

Category	Component	Version	Purpose
Operating System	Windows/linux	Windows 10 / Ubuntu 20.04+	Provides environment for system execution
Programming Language	TypeScript and Python	3.8 or above	Core language for system development
Computer Vision library	OpenCV(CV2)	4.x	Handles video capture, image processing, and visualization
Numerical Library	Numpy	1.x	Performs numerical computations and array operations

Face Detection Model	Retina Face/Face-api.js	Pre-Trained Model	Detects faces and facial landmarks in real time
Face Recognition Framework	InsightFace (ArcFace)	Latest	Generates embeddings and performs face recognition
Deep Learning Backend	TensorFlow/ONNX Runtime	Latest Stable	Execute deep learning model efficiently
Database	SQLite/JSON	Built-in	Stores user data, embeddings, and record attendance
Utility Library	Argparse	Built-in	Handle-command line arguments(optional)

4.1.2 Hardware Requirements

Table 3. Hardware Requirements

Component	Minimum	Recommended	Purpose
Processor	Intel Core i3	Intel Core i5/AMD RYZEN	Executes system operations and processes AI models
RAM	8 GB	16 GB	Supports smooth real-time processing and handling multiple

			faces
Storage	10 GB free space	20 GB or more	Stores application files, models, and attendance data
Camera	Basic Webcam (480p)	HD Webcam (720p or higher)	Captures real-time video input for face detection
Display	Standard Monitor	HD Display	Displays live video feed

4.2 System Implementation

4.2.1 Authentication Screen Implementation

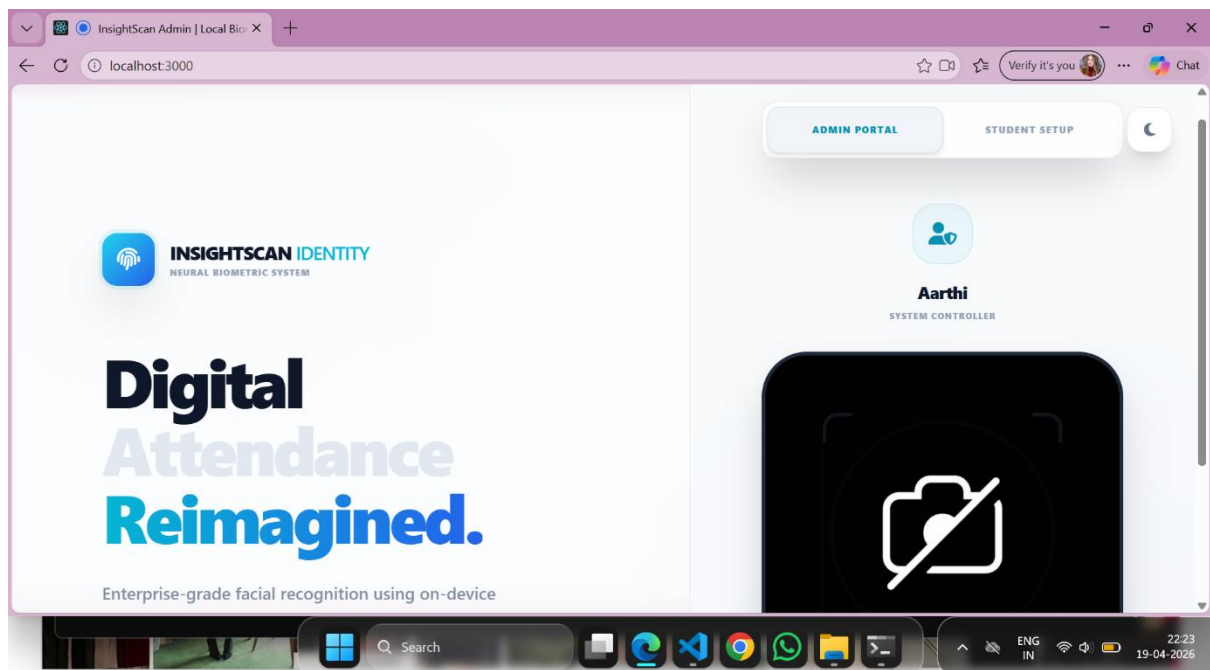


Figure 3. Authentication Screen

The authentication screen acts as the primary entry point of the system and plays a crucial role in ensuring secure access. It supports both administrator login and initial registration functionalities. During the first-time setup, the administrator is required to enter essential details and complete the face enrollment process,

where facial data is captured and stored securely in the system database.

For subsequent logins, the system uses face recognition-based authentication instead of traditional password methods. The webcam captures the admin's face, which is then processed and compared with the stored facial embeddings. If a match is successfully identified, access to the main dashboard is granted; otherwise, access is denied, ensuring system security.

This approach enhances protection against unauthorized access by eliminating the risks associated with password-based systems. Additionally, the interface includes a recovery mechanism, allowing the administrator to resolve login issues or re-register facial data if needed.

4.2.2 Face Scanner Implementation

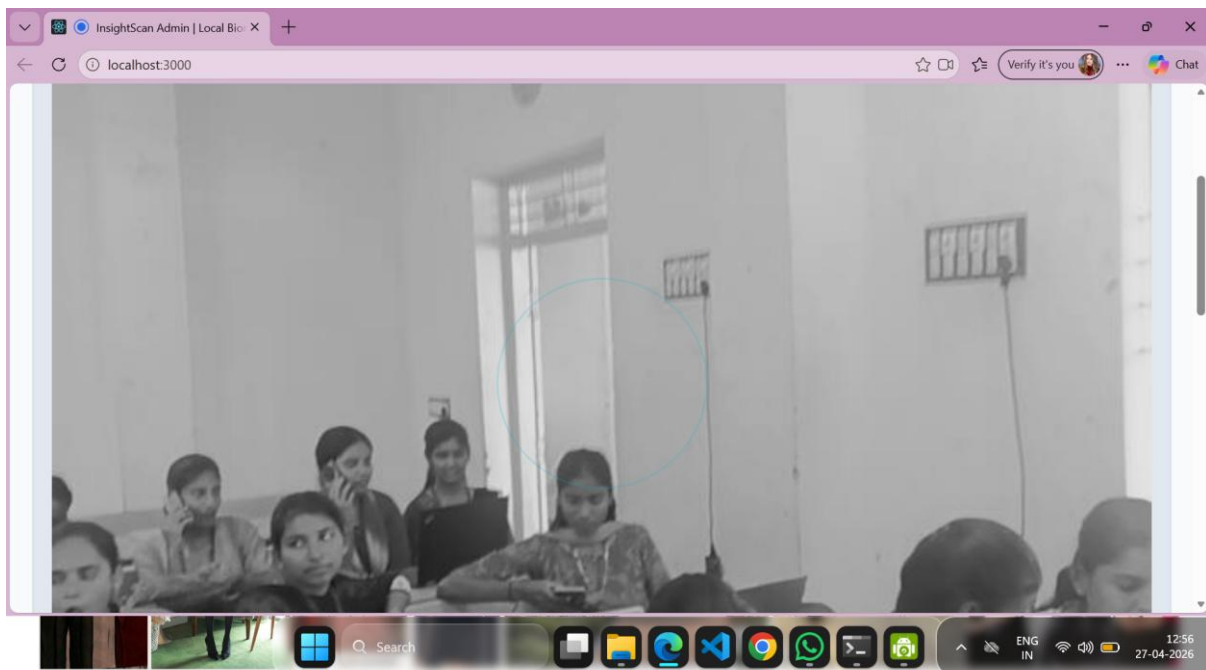


Figure 4. Face Scanner

The face scanner module is responsible for capturing real-time video input through the webcam and detecting human faces within the frame. It continuously processes video frames to identify and locate faces accurately, ensuring reliable input for the recognition system.

Instead of relying on a single image, the module analyzes multiple frames to

achieve stable face capture. This approach helps in reducing errors caused by motion, lighting variations, or unclear images, thereby improving the overall accuracy of the system.

To enhance user interaction, a progress indicator is displayed during the scanning process. This guides the user to maintain proper positioning until sufficient data is captured, ensuring consistent and high-quality face detection.

4.2.3 Student Enrollment Implementation

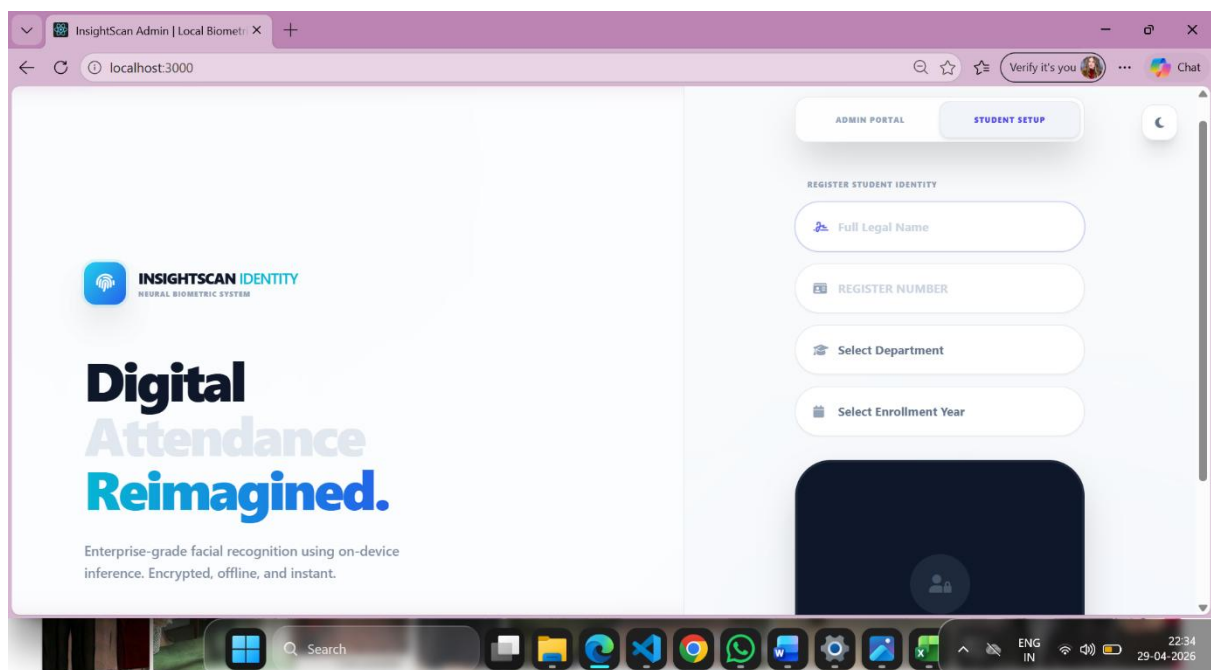


Figure 5. Student Enrollment

The student enrollment module enables the administrator to register new students into the system by entering essential details such as name and registration number. This module ensures that all required information is collected in a structured manner before proceeding with face registration.

During the enrollment process, the system captures the student's facial data using the webcam and generates a unique facial embedding. This embedding acts as a digital representation of the student's face, which will be used later for accurate recognition.

To maintain data integrity, the system checks for duplicate entries before storing the information in the database. Once the registration is successfully completed, a confirmation message is displayed, indicating that the student has been securely enrolled in the system.

4.2.4 Dashboard Implementation

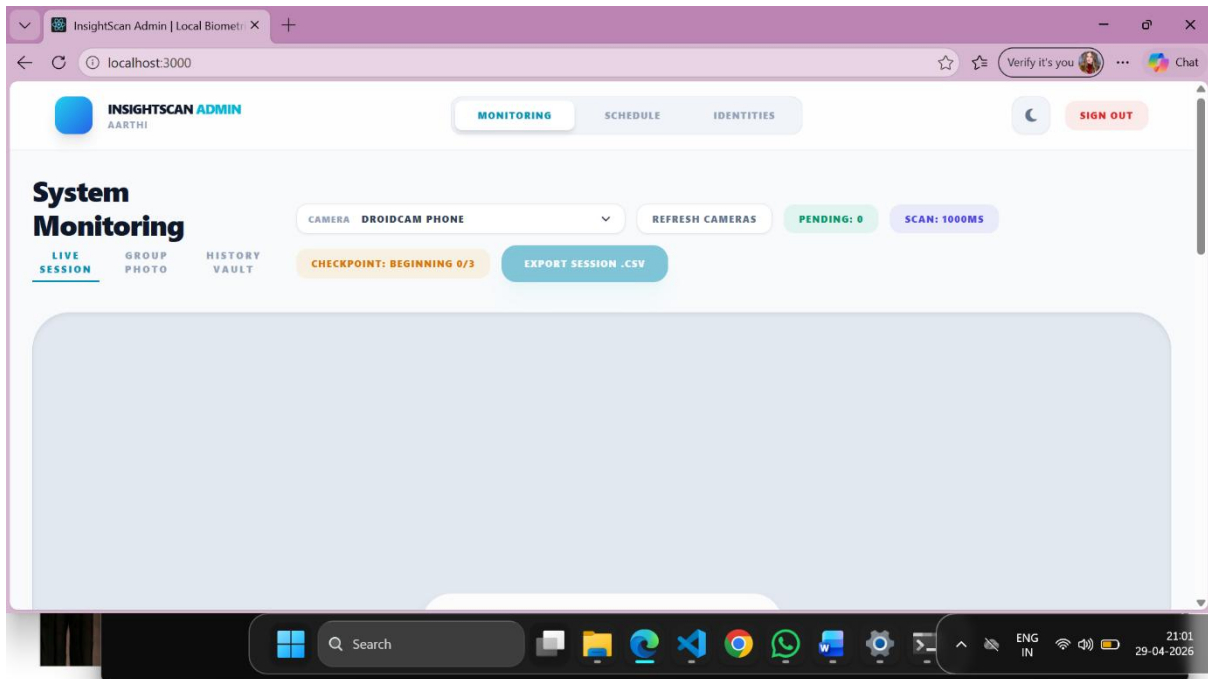


Figure 6. Dashboard

The dashboard serves as the central interface of the system, providing a unified platform for accessing all major functionalities. It acts as the control panel where the administrator can efficiently manage and monitor various system operations in real time.

It provides access to key features such as live attendance monitoring, schedule management, and student records. These features are organized in a structured manner, allowing the user to quickly navigate between different modules and perform tasks without confusion or delay.

The interface is designed to be user-friendly, with clearly defined sections for smooth interaction. Additionally, it supports theme switching, enabling users to

customize the visual appearance for better usability. This enhances the overall user experience and makes the system more convenient for daily use.

4.2.5 Schedule Management Implementation

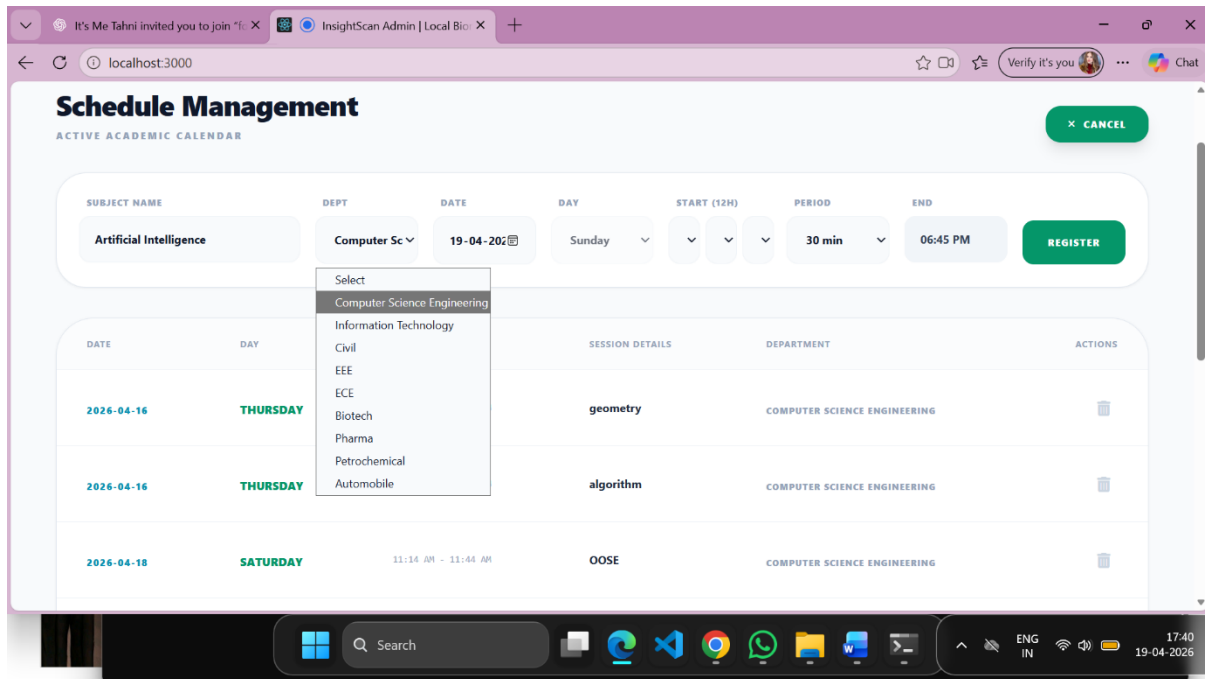


Figure 7. Schedule Management

The schedule management module allows the administrator to create and manage class schedules efficiently within the system. It provides a well-structured interface where the admin can enter all necessary details required for organizing sessions in an orderly manner.

The user inputs information such as subject name, class timing, and duration. Based on these inputs, the system automatically calculates and organizes the schedule, reducing manual effort and minimizing the chances of errors. This automation helps in maintaining consistency in scheduling.

The created schedules are stored in the system and are actively used for attendance tracking during live sessions. This module ensures that attendance is recorded only during valid time slots, improving accuracy. Overall, it helps in maintaining proper class organization, better time management, and smooth system operation.

4.2.6 Live Monitoring Implementation

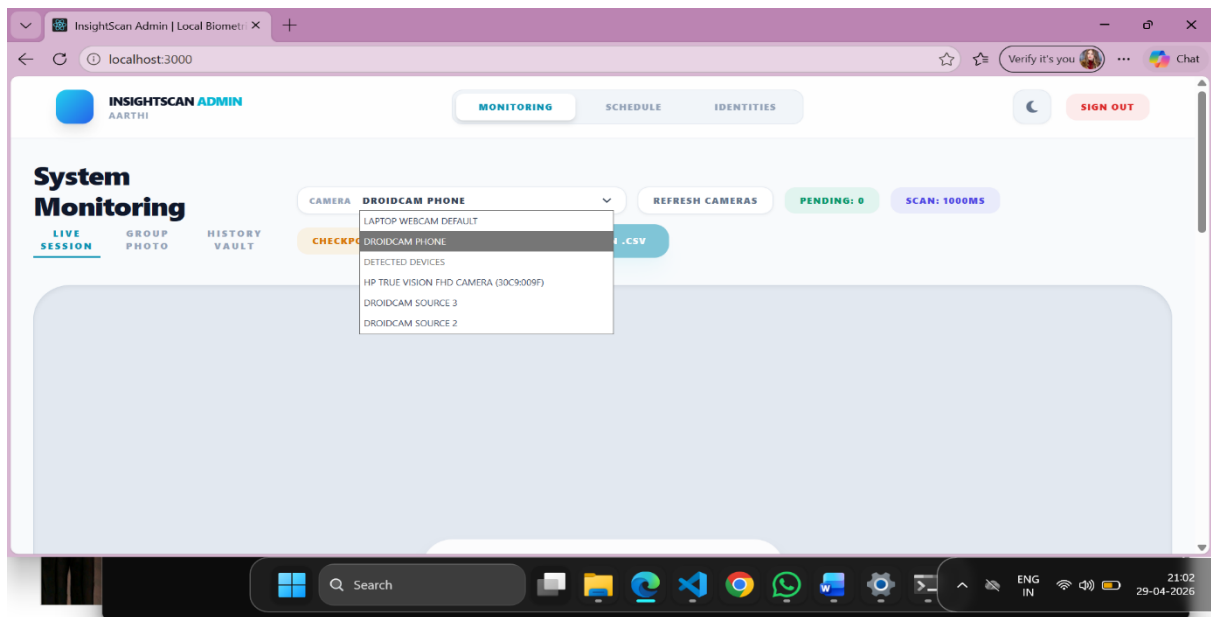


Figure 8(i) Live Monitoring

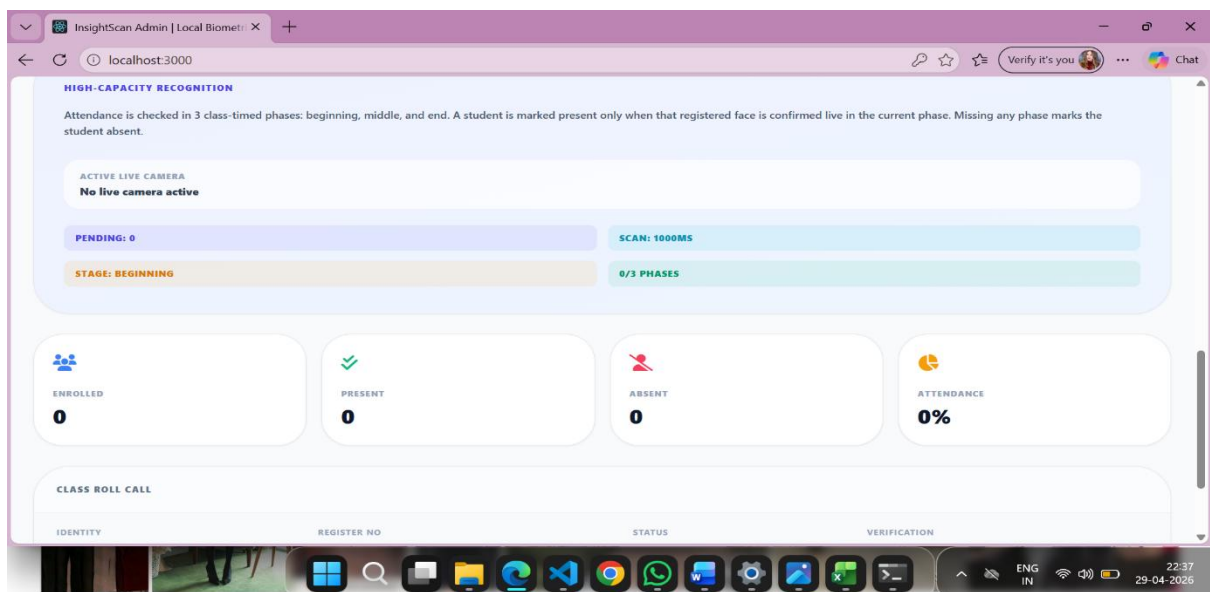
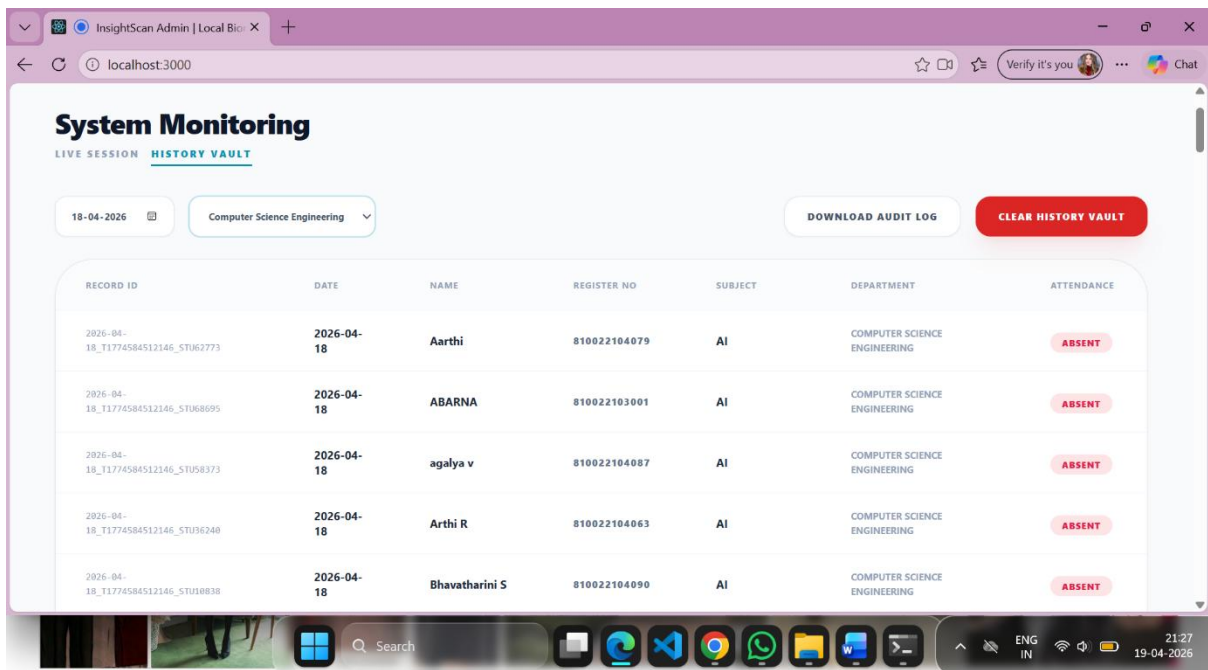


Figure 8(ii) Live Monitoring

The live monitoring module performs real-time face recognition during active class sessions. It detects faces, compares them with stored data, and identifies students present. Attendance is marked only after consistent recognition to ensure accuracy and avoid false entries.

4.2.7 History and Export Implementation



The screenshot displays the 'System Monitoring' interface, specifically the 'HISTORY VAULT' tab. It shows a table of attendance records for 'Computer Science Engineering' on '18-04-2026'. The table includes columns for Record ID, Date, Name, Register No, Subject, Department, and Attendance. Five records are listed, all marked as 'ABSENT'.

RECORD ID	DATE	NAME	REGISTER NO	SUBJECT	DEPARTMENT	ATTENDANCE
2026-04-18_T1774584512146_STU62773	2026-04-18	Aarthi	810022104079	AI	COMPUTER SCIENCE ENGINEERING	ABSENT
2026-04-18_T1774584512146_STU68695	2026-04-18	ABARNA	810022103001	AI	COMPUTER SCIENCE ENGINEERING	ABSENT
2026-04-18_T1774584512146_STU58373	2026-04-18	agalya v	810022104087	AI	COMPUTER SCIENCE ENGINEERING	ABSENT
2026-04-18_T1774584512146_STU96240	2026-04-18	Arthi R	810022104063	AI	COMPUTER SCIENCE ENGINEERING	ABSENT
2026-04-18_T1774584512146_STU10838	2026-04-18	Bhavatharini S	810022104090	AI	COMPUTER SCIENCE ENGINEERING	ABSENT

Figure 9. History and Export

The history module is responsible for storing and managing past attendance records in a well-organized manner. It maintains a complete log of attendance data, including details such as student name, date, time, and session information, ensuring accurate and reliable record keeping over time.

Users can easily view and filter these records based on specific criteria such as date, department, or class. This filtering capability allows quick access to required information without manually searching through large amounts of data, making the system more efficient and user-friendly for administrators.

Additionally, the module provides an export feature that allows attendance data to be downloaded in CSV format. This enables easy integration with external applications such as spreadsheets for analysis, reporting, and documentation. Overall, the history module improves data accessibility, simplifies record management, and supports better decision-making.

4.2.8 Database Implementation

Record ID	Date	Name	Register N	Subject	Departme	Attendanc	Verified C	Verification Timeline	Session Time
2026-04-27_T177	27-04-2026	Aarthi	8.10E+11	cn	Computer Absent	0/3	No checkpoints verified		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	agalya v	8.10E+11	cn	Computer Absent	0/3	No checkpoints verified		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Bhavathar	8.10E+11	cn	Computer Absent	01-Mar	Beginning: 12:48:59 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Gayathri K	8.10E+11	cn	Computer Present	03-Mar	Beginning: 12:46:47 pm Middle: 12:50:32 pm End: 12:53:10 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	hai	8.10E+11	cn	Computer Present	03-Mar	Beginning: 12:48:24 pm Middle: 12:51:41 pm End: 12:53:18 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	jancy	8.1E+08	cn	Computer Present	03-Mar	Beginning: 12:46:44 pm Middle: 12:50:18 pm End: 12:54:02 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Nimitha S	8.10E+12	cn	Computer Present	03-Mar	Beginning: 12:47:26 pm Middle: 12:51:15 pm End: 12:53:47 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	P Preetha	8.10E+11	cn	Computer Present	03-Mar	Beginning: 12:47:33 pm Middle: 12:53:01 pm End: 12:54:50 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Pathmapri	8.10E+11	cn	Computer Present	03-Mar	Beginning: 12:46:28 pm Middle: 12:50:18 pm End: 12:54:34 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Purnima G	8.10E+11	cn	Computer Present	03-Mar	Beginning: 12:47:06 pm Middle: 12:50:55 pm End: 12:55:21 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Sangeetha	8.10E+11	cn	Computer Present	03-Mar	Beginning: 12:47:50 pm Middle: 12:51:27 pm End: 12:53:52 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	sneha	8.1E+10	cn	Computer Present	03-Mar	Beginning: 12:48:09 pm Middle: 12:51:54 pm End: 12:53:35 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Sudeshha N	8.10E+12	cn	Computer Present	03-Mar	Beginning: 12:48:32 pm Middle: 12:51:51 pm End: 12:53:33 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	Swathi G	8.1E+10	cn	Computer Present	03-Mar	Beginning: 12:46:28 pm Middle: 12:52:46 pm End: 12:55:07 pm		12:46 PM - 12:56 PM
2026-04-27_T177	27-04-2026	swathi R	8.10E+11	cn	Computer Present	03-Mar	Beginning: 12:48:25 pm Middle: 12:52:29 pm End: 12:53:19 pm		12:46 PM - 12:56 PM

Figure 10. Database

The system uses a lightweight database to store student information and attendance records in an efficient and organized manner. It maintains essential details such as student identity, facial embeddings, and attendance logs, ensuring that all data is securely stored and easily accessible when required.

The database organizes data in a structured format, which allows quick insertion, updating, and retrieval of records. This improves the speed of operations such as face recognition matching and attendance marking, thereby supporting smooth real-time performance of the system.

Another key advantage of using a lightweight database is that it operates locally without the need for an external server. This makes the system cost-effective, easy to deploy, and suitable for small to medium-scale applications. It also reduces system complexity while maintaining reliable and consistent data management.

4.3 Working Model

4.3.1 Step-by-Step Execution of the System

The working model of the proposed AI-Based Smart Attendance System follows a systematic sequence of operations to ensure accurate and real-time attendance marking. The execution process is described step by step as follows:

Step 1: System Initialization

Initially, the system is started by loading all required components, including the RetinaFace face detection model, InsightFace (ArcFace) recognition model, and the database containing registered user embeddings. The webcam is also initialized for capturing live video.

Step 2: Real-Time Video Capture

The system activates the webcam to capture a continuous video stream. This stream is divided into individual frames, which are processed sequentially for face detection and recognition.

Step 3: Face Detection

Each frame is passed to the RetinaFace model, which detects all faces present in the frame. The model identifies facial regions and along with key facial landmarks.

Step 4: Face Preprocessing and Alignment

The detected face regions are extracted from the frame and undergo preprocessing. This includes resizing the image to a standard size, normalizing pixel values, and aligning the face using detected landmarks. This step ensures consistency in input data.

Step 5: Feature Extraction

The preprocessed face image is then passed to the InsightFace (ArcFace) model, which generates a 512-dimensional embedding vector. This vector represents the unique facial features of the individual.

Step 6: Face Recognition

The generated embedding is compared with stored embeddings in the database

using cosine similarity. A similarity score is calculated to determine how closely the detected face matches a registered user.

Step 7: Decision Making

Based on the similarity score :If the score is above the predefined threshold, the face is recognized as a registered user .If the score is below the threshold, it will not recognize face .This step ensures accurate identification and prevents false recognition.

Step 8: Attendance Marking

If the face is successfully recognized, the system automatically records attendance. The following details are stored:

Record ID, Date, Name, Register Number , Subject , Department, Attendance, Verified Checkpoint, Verification Timeline ,Session Time.

Step 9: Display Output

The system displays the processed output on the screen, including :Name of recognized individuals and to mark attendance .This provides real-time feedback to the user.

Step 10: Continuous Processing

The system continues to process incoming frames in real time, repeating the above steps for each frame until the program is terminated.

4.4. Results & Outputs

The proposed AI-based smart attendance system using face recognition for classroom was successfully implemented and tested under controlled conditions. The system was able to detect and recognize faces in real time with high accuracy, ensuring reliable attendance marking. The use of deep learning models enabled the system to perform effectively even with variations in lighting, pose, and facial expressions.

4.4.1 Admin Biometric Login

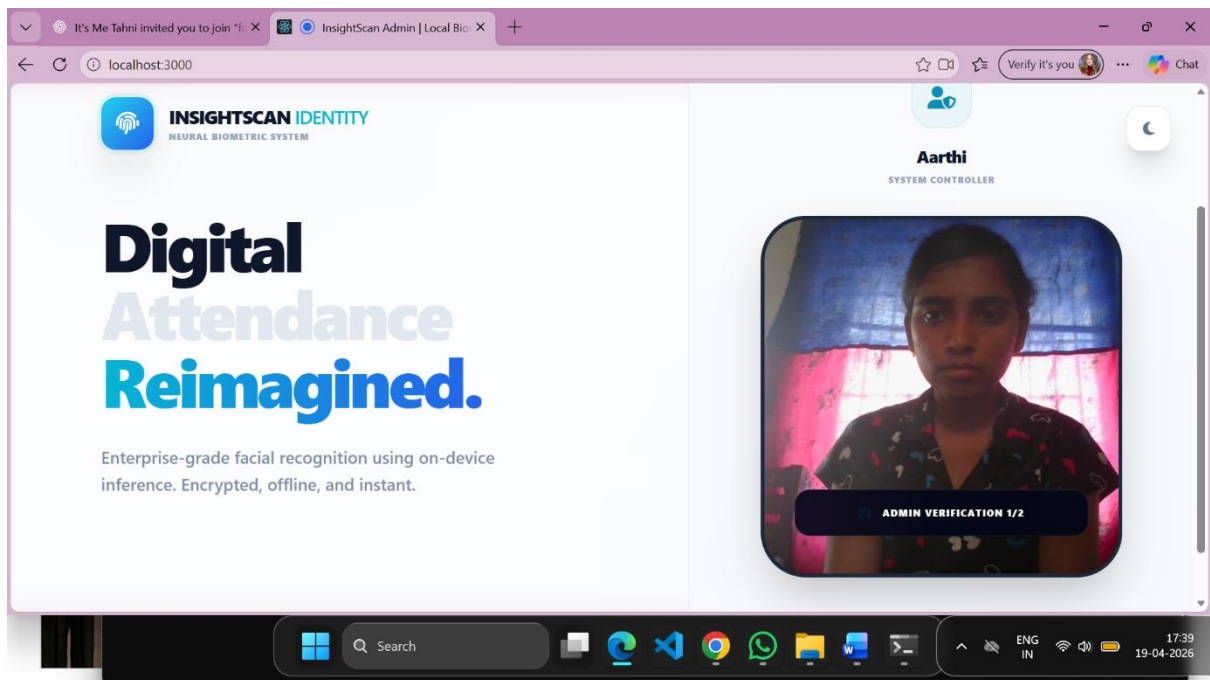


Figure 11. Admin Biometric Login

4.4.2 Student Biometric Registration

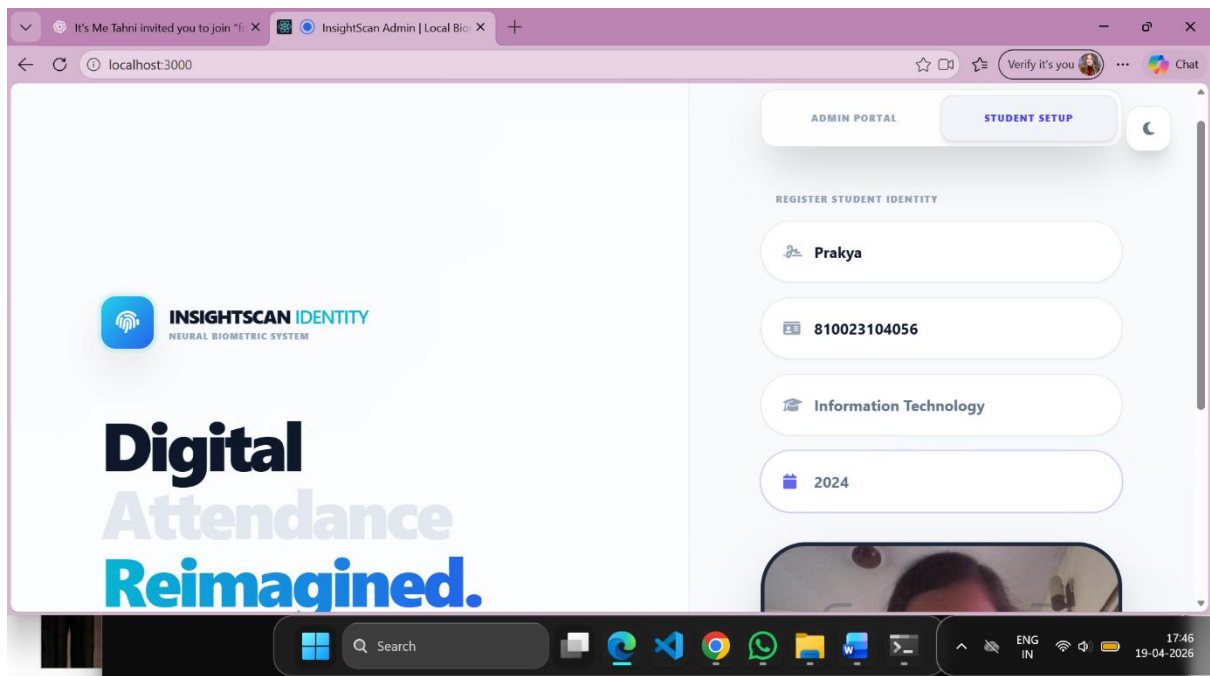


Figure 12. Student Biometric Registration

4.4.3 Schedule Creation

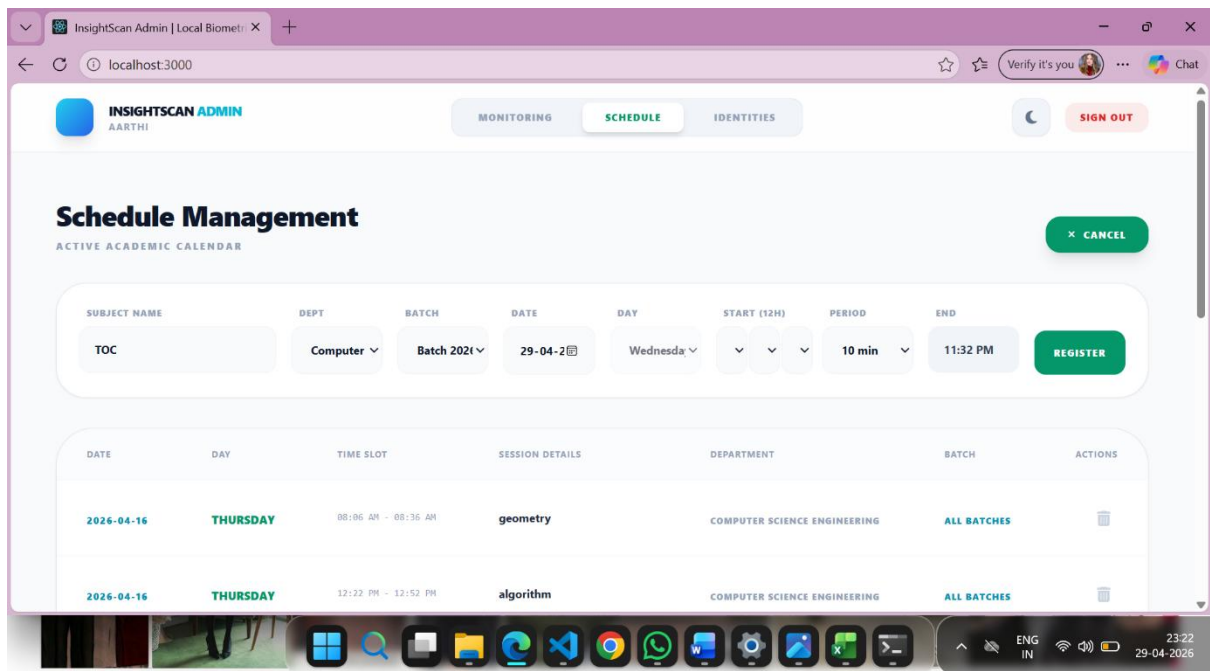


Figure 13. Schedule Creation

4.4.4 Live Attendance Monitoring

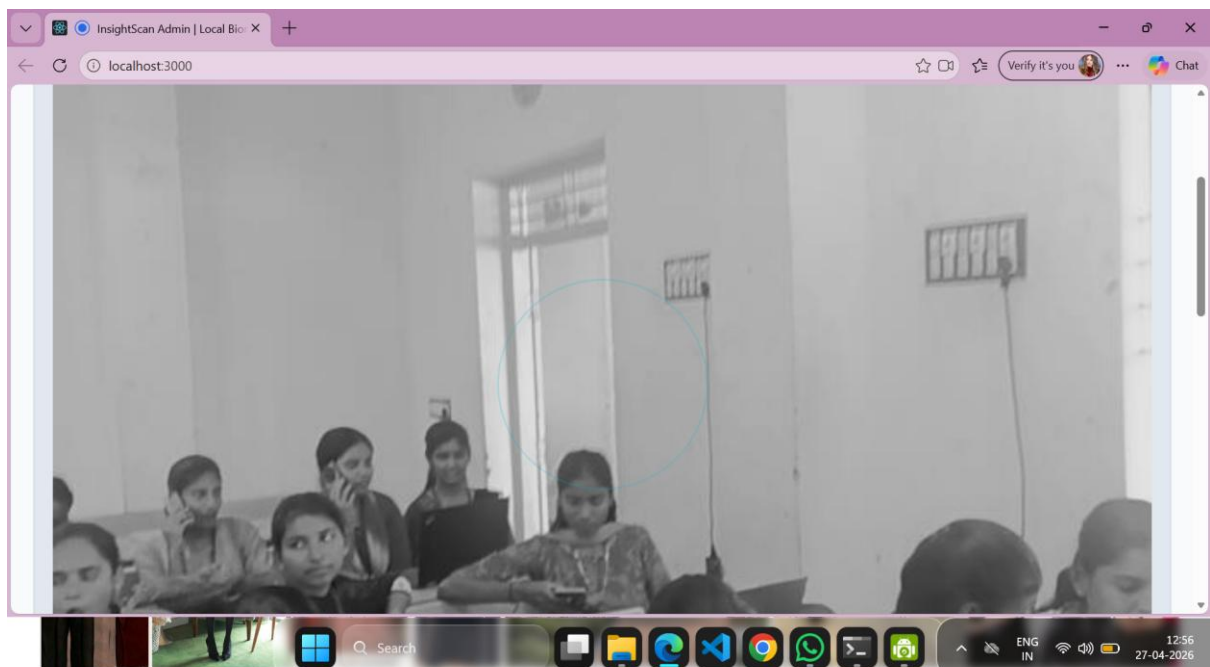


Figure 14. Live Attendance Monitoring

4.4.5 Checkpoint-Based Presence Marking

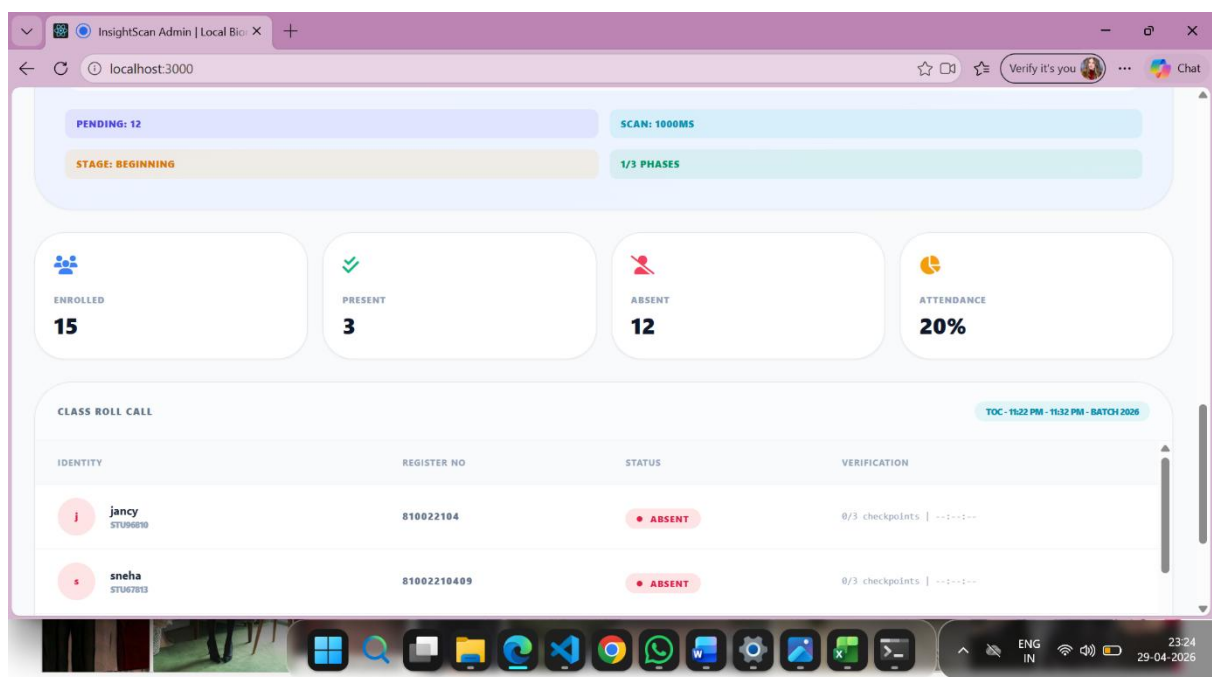


Figure 15. Checkpoint-Based Presence Marking

4.4.6 History Review

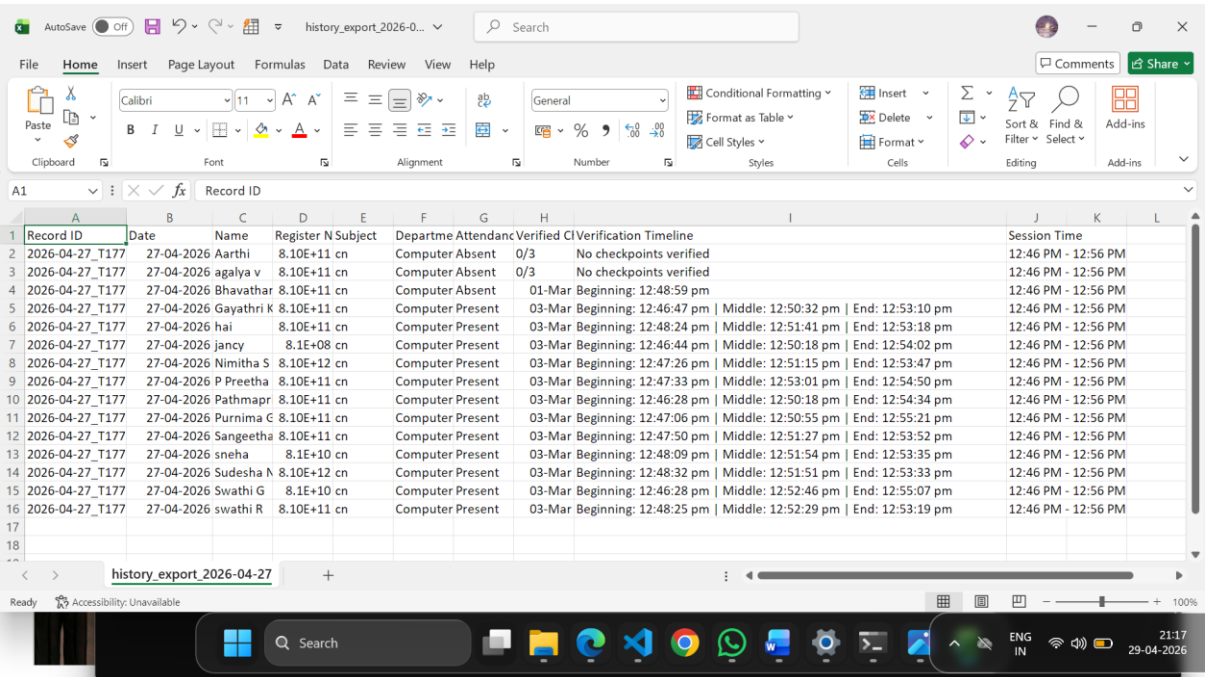


Figure 16. History Review

4.5 Performance Analysis

The performance of the proposed AI-based attendance system was evaluated based on accuracy, speed, and reliability. It performs real-time detection with minimal delay, providing quick and smooth attendance marking. The system can handle multiple faces simultaneously, making it suitable for classroom environments.

4.6.1 Accuracy Analysis

The accuracy of the proposed AI-Based Smart Attendance System is evaluated based on its ability to correctly recognize individuals and reliably mark attendance in real-time conditions. The system utilizes advanced deep learning models such as RetinaFace for face detection and InsightFace (ArcFace) for face recognition, which significantly enhances recognition performance.

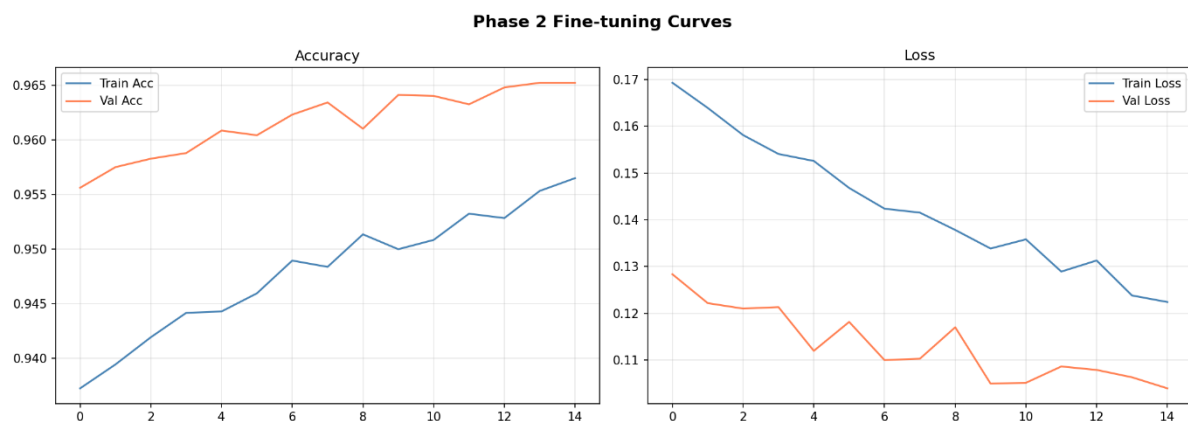


Figure 17. Accuracy Graph

The system achieves high accuracy, approximately above 90% under controlled conditions, due to the use of robust facial feature extraction and matching techniques. Instead of relying on a single image, the system processes multiple frames to generate stable facial embeddings, which reduces noise and improves recognition reliability.

Face matching is performed using cosine similarity, ensuring that only highly similar facial features are accepted as a match. This dual-validation approach minimizes false positives (incorrect matches) and false negatives (missed

matches). Additionally, the system compares the best match with alternative candidates to avoid ambiguity when multiple faces are present.

Attendance accuracy is further improved by validating presence across multiple time intervals (beginning, middle, and end of the session). A student must be consistently detected during these checkpoints to be marked present. This prevents false attendance marking due to temporary or incorrect detections and ensures that attendance reflects actual participation.

Overall, the system provides reliable and consistent performance, making it suitable for real-time classroom environments.

Efficiency Analysis

The efficiency of the system is evaluated in terms of both operational efficiency and computational efficiency.

Operational Efficiency

The proposed system significantly reduces manual effort compared to traditional attendance methods. In conventional systems, teachers spend considerable time marking attendance manually or verifying students individually. In contrast, this system automates the entire process, allowing attendance to be recorded instantly using face recognition.

The system operates only during scheduled sessions, which avoids unnecessary processing outside class hours. This improves overall system efficiency and ensures that attendance is recorded only when required. Additionally, attendance records are automatically stored and can be easily retrieved or exported, reducing administrative workload.

Computational Efficiency

The system is designed to optimize computational performance by reducing unnecessary processing. Only relevant student data is loaded based on the active session, which minimizes the number of comparisons required for face matching.

The system processes frames at controlled intervals, ensuring smooth real-time performance without overloading system resources. Once a student is successfully

recognized and marked present, repeated processing for that individual is avoided, further improving efficiency.

These optimizations enable the system to maintain fast response time and stable performance, even when handling multiple users simultaneously.

Comparison with Existing System

Table 4. Comparison with Existing System

Parameter	Manual System	RFID System	Biometric System	Proposed System
Accuracy	Low(60-70%)	Moderate (75–85%)	High (85-90%)	Very High (95%)
Proxy Attendance	Possible	Possible	Difficult	Eliminated
Time Efficiency	Low	Moderate	Moderate	High(Real-Time)
User Interaction	Manual	Card-based	Contact-based	Contactless
Scalability	Low	Moderate	Moderate	High
Hardware Dependency	None	RFID Devices	Biometric Devices	Minimal (Webcam)

Conclusion & Future Enhancement

5.1 Conclusion

The project titled AI-Based Smart Attendance System Using Face Recognition successfully demonstrates the implementation of an intelligent, automated, and reliable attendance management system using advanced deep learning techniques. The system effectively integrates RetinaFace for face detection and InsightFace (ArcFace) for face recognition, enabling accurate identification of individuals in real time. By utilizing facial embeddings and cosine similarity for matching, the system ensures high precision in recognizing registered users while minimizing false detections.

The developed system eliminates the limitations of traditional attendance methods such as manual errors, proxy attendance, and time inefficiency. It provides a fully automated, contactless, and secure solution, making it highly suitable for modern educational environments. Experimental results show that the system achieves high accuracy (around 95% or above) and performs efficiently in real-time scenarios, even when handling multiple faces simultaneously.

The use of a lightweight database ensures effective storage and retrieval of attendance records. Overall, the proposed system meets its primary objective of automating attendance using face recognition and proves to be a scalable, efficient, and practical solution. This project highlights the potential of Artificial Intelligence in transforming traditional administrative tasks into smart and intelligent systems.

5.2 Future Enhancement

5.2.1 Improvements

The proposed system can be further improved to enhance its accuracy, security, and performance. One major improvement is the use of larger and more diverse datasets to improve recognition accuracy under varying conditions such as lighting, pose variations, and facial occlusions. The system can also be enhanced by implementing advanced model optimization techniques to reduce processing time and improve real-time performance.

Another important improvement is the integration of liveness detection, which helps prevent spoofing attacks using photos or videos. Additionally, continuous learning mechanisms can be introduced so that the system can adapt to changes in user appearance over time.

The system can also be optimized by improving hardware utilization, such as leveraging GPU acceleration for faster processing. Enhancing the user interface to make it more interactive and user-friendly is another area of improvement.

5.2.2 Real-World Extensions

The system can be extended for real-world applications across various domains beyond classroom attendance. It can be deployed in corporate offices for employee attendance tracking and access control. The system can also be integrated into smart campus environments, where attendance data is automatically synchronized with institutional databases.

Another important extension is the development of a mobile-based application, allowing attendance monitoring through smartphones. Integration with cloud platforms can enable centralized data storage and remote access from multiple locations.

The system can also be expanded to include additional features such as emotion recognition, behaviour analysis, and analytics dashboards, which can provide insights into user engagement and attendance trends.

APPENDICE

```
from flask import Flask, request, jsonify, render_template
import sqlite3
import face_recognition
import numpy as np
import base64
import cv2
app = Flask(__name__)
DB = "database.db"
# ----- DB SETUP -----
def init_db():
    conn = sqlite3.connect(DB)
    c = conn.cursor()
    c.execute("CREATE TABLE IF NOT EXISTS admin (
        id INTEGER PRIMARY KEY,
        name TEXT,
        face BLOB,
        password TEXT,
        email TEXT
    )")
    c.execute("CREATE TABLE IF NOT EXISTS students (
        id INTEGER PRIMARY KEY,
        name TEXT,
        reg_no TEXT,
        dept TEXT,
        year INTEGER,
        face BLOB
    )")
    conn.commit()
```

```

        conn.close()
init_db()
# ----- UTIL -----
def decode_image(base64_img):
    img_data = base64.b64decode(base64_img.split(",")[1])
    np_arr = np.frombuffer(img_data, np.uint8)
    return cv2.imdecode(np_arr, cv2.IMREAD_COLOR)
def get_face_encoding(image):
    rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    encodings = face_recognition.face_encodings(rgb)
    return encodings[0] if encodings else None
# ----- ADMIN -----
@app.route("/register_admin", methods=["POST"])
def register_admin():
    data = request.json
    image = decode_image(data["image"])
    encoding = get_face_encoding(image)
    if encoding is None:
        return jsonify({"error": "No face detected"})
    conn = sqlite3.connect(DB)
    c = conn.cursor()
    c.execute("DELETE FROM admin")
    c.execute("INSERT INTO admin (name, face, password, email) VALUES
(?, ?, ?, ?)",
        (data["name"], encoding.tobytes(), data["password"], data["email"]))
    conn.commit()
    conn.close()
    return jsonify({"message": "Admin registered"})
@app.route("/verify_admin", methods=["POST"])

```

```

def verify_admin():
    image = decode_image(request.json["image"])
    encoding = get_face_encoding(image)
    if encoding is None:
        return jsonify({"error": "No face detected"})
    conn = sqlite3.connect(DB)
    c = conn.cursor()
    c.execute("SELECT face FROM admin LIMIT 1")
    row = c.fetchone()
    conn.close()
    if not row:
        return jsonify({"error": "No admin found"})
    stored = np.frombuffer(row[0], dtype=np.float64)
    match = face_recognition.compare_faces([stored], encoding,
tolerance=0.5)[0]
    return jsonify({"verified": bool(match)})
# ----- STUDENT -----
@app.route("/register_student", methods=["POST"])
def register_student():
    data = request.json
    image = decode_image(data["image"])
    encoding = get_face_encoding(image)

    if encoding is None:
        return jsonify({"error": "No face detected"})
    conn = sqlite3.connect(DB)
    c = conn.cursor()
    # check duplicate face
    c.execute("SELECT face FROM students")

```

```

for (face_blob,) in c.fetchall():
    existing = np.frombuffer(face_blob, dtype=np.float64)
    match = face_recognition.compare_faces([existing], encoding,
tolerance=0.5)[0]
    if match:
        return jsonify({"error": "Duplicate face detected"})
    # check duplicate reg no
    c.execute("SELECT * FROM students WHERE reg_no=?",
(data["reg_no"],))
    if c.fetchone():
        return jsonify({"error": "Register number exists"})
    c.execute("INSERT INTO students (name, reg_no, dept, year, face)
VALUES (?, ?, ?, ?, ?)",
        (data["name"], data["reg_no"], data["dept"], data["year"],
encoding.tobytes()))
    conn.commit()
    conn.close()
    return jsonify({"message": "Student registered"})
# ----- UI -----
@app.route("/")
def home():
    return render_template("index.html")
if __name__ == "__main__":
    app.run(debug=True)

```

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